

Preventive Health

Unomi: Mungkut Toothpaste

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: Mae Fah Luang University





Key Highlights

- Mangosteen xanthone extract combined with Korean bamboo salt for oral health
- Proven antibacterial activity against Streptococcus mutans, the primary cause of tooth decay



Mangosteen xanthone extract



Antibacterial activity against Streptococcus mutans



13 bioactive compounds

Scientific testing also identified 13 bioactive compounds, strengthening the product's claims, credibility, and market potential as an FDA-notified herbal toothpaste.

More Information

RECAL: High-Protein, Complete-Formula Meal Replacement for Weight Control or Weight Loss

Mr.Phuwis lewsipo
Affiliated Organization: **Nutrition Experts Co., Ltd.**

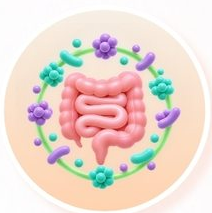


Key Highlights

- Low-calorie, high-protein complete meal replacement for weight control
- Stimulates the body's own GLP-1 hormone to regulate appetite and reduce cravings



Low-calorie, high-protein formula



Helps extend satiety



Dietary fiber and beneficial fatty acids

Dual protein blend:
high-quality milk and plant proteins help preserve lean muscle, maximize satiety, and provide a balanced amino acid profile.

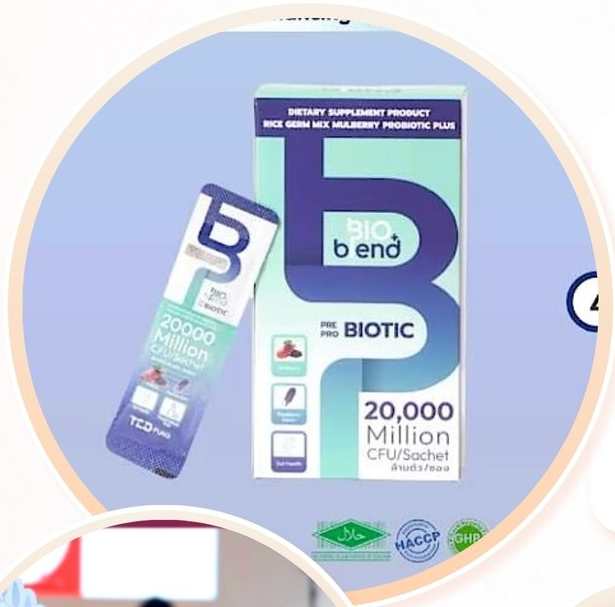
Expandable fiber and healthy fats:
highly expandable dietary fiber and beneficial fatty acids delay gastric emptying and prolong fullness.

Sequential weight management:
a structured formula that optimizes each phase of digestion for sustainable, effective weight loss.

More Information

BioBlend Plus: Thai Probiotics Enriched with Mulberry and Riceberry Germ Extracts

Assoc. Prof. Dr.Doungporn Amornlerdpison
Affiliated Organization: Agri Inno, Maejo University





Key Highlights

- Synbiotic formula of selected probiotics with prebiotics from mulberry fruit and riceberry germ
- Supports digestion, bowel regularity, and immunity, and helps reduce gastrointestinal infections
- Validated by metagenomic and metabolomic studies

Probiotics and prebiotics

Gut microbiome balance

Supports digestion and immunity

How it works: promotes the growth and activity of beneficial gut microorganisms, producing essential bioactive metabolites that maintain a balanced gut microbiome.

Scientific evidence: metagenomic studies confirmed increased diversity and population of beneficial gut bacteria, while metabolomic analysis identified and quantified the key bioactive compounds behind these effects.

More Information

Next-generation wellness: integrates biotechnology and nutrition science to deliver a safe, natural, evidence-based solution for long-term gut health.

Coffogenic Drink: Dietary Supplement Coffee Pulp Extract

Naruemon Taksa-Udom
Affiliated Organization: Hillkoff Co., Ltd. and Chiang Mai University







Key Highlights

- Upcycles coffee husks, an agro-industrial by-product, into a premium functional drink
- Clinically shown to reduce LDL, triglycerides, total cholesterol, insulin resistance, and inflammation while increasing HDL



Upcycled coffee fruit



Human clinical research



Supports healthy blood lipid management

Clinical study group: overweight individuals with high blood lipids (dyslipidemia).

Circular economy and low-carbon model: adds value to coffee waste, reduces environmental burden, and strengthens Hillkoff's low-carbon coffee value chain.

Sustainable community impact: supports highland farmers with stable income opportunities and promotes forest-friendly coffee cultivation in Northern Thailand.

More Information

ImuneUp: Dietary Supplement from Golden Cordyceps in Oral Powder Form

Wayuree Sukrungruangsanti
Affiliated Organization: **Bist Inno Reform Co., Ltd.**







Key Highlights

- Golden Cordyceps mycelium + 8 probiotic strains + Vitamins C, B6, and B12 for gut and immune wellness
- Clinical studies: 8 weeks of use enhanced immune responses, reduced allergy symptoms, and lowered inflammatory mediators
- Ready-to-take powder shot, no water needed



Golden Cordyceps



8 probiotic strains



Ready-to-take powder shot

Clean formula: no added sugar and no milk ingredients, suitable for health-conscious consumers and those with lactose sensitivity.

Gut-immune connection: the probiotic strains are selected to support gut microbial balance, working with Golden Cordyceps to link digestive balance and immune defense in one convenient daily format.

More Information

Nutriflow: Low-GI Meal Replacement for Blood Sugar Control

Assoc. Prof. Dr.Thunnalin Winuprasith
Affiliated Organization: Mahidol University



Key Highlights

- Low-GI (< 50) meal replacement with digestion-resistant complex carbohydrates for blood sugar management
- Adjusts to 4 IDDSI texture levels from a single sachet

GI below 50



Adaptive Flow Technology



4 IDDSI texture levels

Adaptive Flow Technology: proprietary, particle-engineered natural hydrocolloids enhance dispersion and water retention, ensuring stability with no phase separation.

Dysphagia and nutritional care: customizable consistency safely meets different swallowing difficulties and individual nutritional needs.

Next-generation meal replacement: combines clinical nutrition science with fluid dynamics technology, offering a practical, safe, and flexible solution for patient and elderly care.

More Information

Modgut: Gut Microbiome Testing for Personalized Weight Management

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: King Mongkut's University of Technology Thonburi & Mod Gut Co., Ltd.





Key Highlights

- Gut microbiome testing for personalized nutrition and weight management
- 16S rRNA sequencing identifies and quantifies gut microorganisms at the genus level



16S rRNA Sequencing



Gut microbiome analysis



Personalized nutrition

Why the gut microbiome: unlike genetic information, which remains largely unchanged, the gut microbiome responds to diet, lifestyle, and behavior, making it a dynamic tool for designing and monitoring personalized nutrition.

Actionable guidance: translates microbial data into practical, individualized nutrition and weight-management recommendations.

More Information

Evidence-based platform: combines molecular diagnostics with nutrition science as an ongoing monitoring tool for long-term wellness and obesity prevention.

Nutrical: Complete Medical Nutrition Formula for Individuals with Diabetes

Asst. Prof. Dr.Supat Chaiyakul
Affiliated Organization: Mahidol University



Key Highlights

- Lactose-free complete nutrition formula for people with diabetes
- Clinically proven low GI (27.86) and GL (16)
- 90-day study: significant improvements in blood glucose, insulin, and triglycerides



Lactose-free formula



GI 27.86



Medical nutrition for diabetes

Nutrition profile: about 1.06–1.07 kcal/mL, with 4.67% protein, 3.73% total fat (1.42% saturated), and 12.09% carbohydrates, within recommended guidelines for diabetic patients.

Complete nutrients: essential amino acids, omega-3, -6, and -9 fatty acids, vitamins, minerals, and dietary fiber in line with established nutritional standards.

Additional clinical results: reductions in systolic blood pressure, body weight, BMI, waist circumference, body fat percentage, and adipose tissue proportion.

Who it's for: people with diabetes or lactose intolerance, recovering patients, the elderly, and anyone needing balanced, safe nutritional supplementation.

More Information

Albuminuria Test Kit for Early Chronic Kidney Disease Screening

Prof.Nattachai Srisawat, M.D., Ph.D.
Affiliated Organization: Chulalongkorn University



Key Highlights

- Simple self-screening urine test that detects early microalbuminuria – a key sign of reduced kidney function – before symptoms appear
- Easy to use and easy to read, with high performance and accuracy
- No hospital visit or travel needed for initial screening



Urine microalbumin screening



Self-testing



Supports early risk detection

Why it matters: early-stage chronic kidney disease often develops without obvious symptoms, and conventional screening requires a hospital visit, adding time and travel costs – especially for people living far from healthcare facilities.

Integrated digital health platform: combines certified health-monitoring devices, mobile connectivity, AI-based health analytics, cloud infrastructure, and telemedicine into a single scalable platform.

More Information

Preventive health awareness: supports earlier awareness and timely follow-up when abnormal results are detected, encouraging proactive lifestyle adjustments.

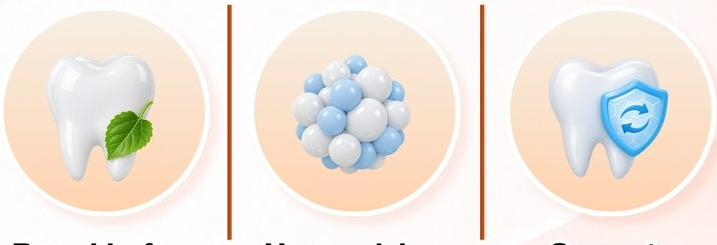
Olikap™ Mulberry Whitening Gel: Peroxide-free Whitening with Nano Calcium Phosphate

Asst. Prof. Dr.Khrongkhwan Akkarachaneeyakorn
Affiliated Organization: Kasetsart University



Key Highlights

- Peroxide-free tooth whitening with mulberry-derived actives
- Nano calcium phosphate supports enamel remineralization
- 3-in-1 care: whitening, enamel repair, and sensitivity reduction



Peroxide-free whitening | **Nano calcium phosphate** | **Supports enamel remineralization**

How it works: nano calcium phosphate particles release calcium and phosphate ions to promote natural enamel remineralization.

Sensitivity care: helps occlude exposed dentinal tubules, a key mechanism for reducing tooth sensitivity.

Gentle at-home use: avoids the harshness of conventional peroxides and is designed for safe, convenient home oral care.

More Information

Nanozeree: Synbiotic Gummy Dietary Supplement Product

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: Prince of Songkla University







Key Highlights

- Robotic Nanoencapsulation protects probiotics from stomach acid and delivers them precisely to the intestine
- About 1 billion CFU of Danish probiotic strains: Lactobacillus rhamnosus and Bifidobacterium animalis
- Turmeric extract with pure curcuminoids, 4x more effective as an antioxidant



Synbiotic gummy



Around 1 billion CFU probiotics



Robotic Nanoencapsulation

Optimal dose: the 1 billion CFU level is backed by research as neither too little (subdose) nor too much (overdose).

Curcuminoid benefits: studies show curcuminoids help reduce digestive inflammation, cope with leaky gut syndrome, reduce cancer formation, and help prevent osteoclastogenesis.

Gut and immune support: research shows these probiotics strengthen immunity, help cope with irritable bowel syndrome, reduce digestive inflammation linked to food allergy, and improve nutrient absorption.

More Information

Mind Kombucha POP: Synbiotic Kombucha Powder

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: Mae Fah Luang University

Key Highlights

- Sugar-free synbiotic powder for everyday digestive wellness
- More than 20 billion CFU of two selected Lactobacillus strains per sachet
- Complete formula: probiotics, prebiotics, and postbiotics



Sugar-free



Over 20 billion CFU per sachet



Kombucha, mulberry, and inulin

30-day fermented kombucha: contains components from green tea kombucha fermented for 30 days, providing natural organic acids, polyphenols, and active metabolites.

Botanical and nutrient complex: mulberry powder, inulin prebiotic fiber, and L-ascorbic acid (Vitamin C) support gut microbiome balance and antioxidant protection.

Digestive support: promotes nutrient absorption, intestinal barrier function, and long-term digestive balance.

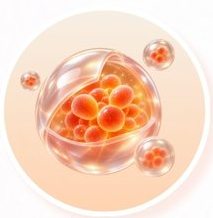
More Information

Songkhla Mini Mango : Probiotic Supplement

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: Mae Fah Luang University

Key Highlights

- Microencapsulation protects live probiotics from heat, moisture, gastric acid, and bile salts
- Delivers probiotics to the colon for gut microbial balance
- Flavored with Songkhla Mini Mango for a uniquely Thai identity



Microencapsulation technology



Helps protect probiotics



Songkhla Mini Mango flavor

Songkhla Mini Mango is a distinctive local fruit from Songkhla Province. By turning this regional fruit into a functional wellness product, the formulation combines microbiome science with local flavor to support daily digestive wellness.

More Information

HARUNA: Natural Prebiotic

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: VTK Inno Group Co., Ltd.



Key Highlights

- 100% natural prebiotic vegetable powder with resistant starch technology
- Supports 9 key groups of beneficial gut microorganisms
- Boosts production of short-chain fatty acids (SCFAs)



100% natural prebiotic powder



Resistant starch technology



Supports beneficial gut microorganisms

How it works: the resistant starch passes through the stomach and small intestine undigested and reaches the colon, where it feeds beneficial gut microorganisms.

Why SCFAs matter: these metabolites support a balanced intestinal environment, intestinal barrier protection, and digestive regularity.

3P concept: Prebiotic → Probiotic Activity → Postbiotic, for a healthier and more balanced gut ecosystem.

More Information

Elite Smile: Postbiotic Toothpaste

Prof. Dr.Rawee Teanpaisan
Affiliated Organization: Prince of Songkla University



Key Highlights

- Postbiotic toothpaste with derivatives from Lactobacillus paracasei SD1 and SD11
- More than 12 natural herbal ingredients
- Helps maintain a balanced oral microbiome



Postbiotic technology



More than 12 botanical ingredients



Supports oral microbiome balance

Herbal blend: includes centella asiatica, mangosteen peel extract, green tea, chamomile, and more, to soothe gums and provide natural antibacterial protection.

Gum support: helps reduce oral inflammation and plaque buildup for long-term gum health.

Long-lasting freshness: postbiotic metabolites and natural botanicals help control bad breath throughout the day.

Gentle daily formula: a non-abrasive toothpaste that bridges microbiome biotechnology with traditional herbal ingredients.

More Information



Medical innovation

Plai Nano Emulsion Spray for Muscle and Joint Relief

Assoc. Prof. Dr.Somsak Nualkaew
Affiliated Organization: Mahasarakham University







Key Highlights

- Optimized Plai extract rich in anti-inflammatory compounds
- Nanoemulsion formula for better skin penetration
- Relieves muscle and joint inflammation



Anti-inflammatory Plai extract



Enhanced skin penetration



Relieves pain and inflammation

An accessible, cost-effective herbal alternative that also creates value for producers, local farmers, and Thailand's herbal industry.

More Information

Yataprasen

Affiliated Organization: **Department of Thai Traditional and Alternative Medicine**







Key Highlights

- Modernized Thai traditional medicine: the traditional Yataprasen formula reinvented as an innovative film-forming spray, bridging Thai wisdom with modern pharmaceutical science
- Comparable to standard pain-relief gel: non-inferior efficacy to Diclofenac gel (a standard NSAID) for musculoskeletal pain and inflammation
- Clinically proven safe: no skin irritation in healthy volunteers, confirming its safety for topical use



Film-forming spray



Clinically tested for safety



Comparable efficacy to standard gel

Multi-target anti-inflammatory action: scientifically validated to reduce key inflammatory mediators, with antioxidant and anti-inflammatory benefits for pain relief.

Better usability and stability: film-forming technology improves convenience, adherence, and bioavailability, keeping the product stable for up to 6 months.

More Information

High commercial and policy potential: ready for herbal product registration and scalable production, supporting evidence-based Thai Traditional Medicine and reducing reliance on imported synthetic pain relievers.

Amrukawati Throat Spray

Affiliated Organization: **Department of Thai Traditional and Alternative Medicine**



Key Highlights

- Modernized Thai traditional medicine: a long-established Thai traditional formula transformed into a convenient, modern throat spray
- Targeted, fast-acting relief: delivers active compounds directly to the throat and upper respiratory tract to ease cough, phlegm, and throat irritation



Modern Thai traditional medicine



Standardized herbal extract



Relieves cough, phlegm, and throat irritation

Standardized herbal extraction: modern extraction technology concentrates key herbal compounds and standardizes active ingredients for consistent quality and industrial-scale production.

Value-added herbal innovation: bridges Thai traditional wisdom with modern pharmaceutical technology, driving the future of the high-value herbal medicine industry.

More Information

SI-HERB DRESSING : Non-Adherent Herbal Lipido-Colloid Wound Dressing

Affiliated Organization: **Mahidol University and Bangkok Botanica Co., Ltd.**







Key Highlights

- Non-adherent herbal wound dressing that reduces pain and protects new tissue during removal
- Clinically supports epithelialization and tissue regeneration in burn and scald patients
- Extends dressing changes from daily to approximately every 2-5 days



Non-adherent and pain-reducing



Supports a moist wound environment



Centella asiatica and Aloe vera

Advanced structure: a flexible polyester mesh coated with lipido-colloid particles and enriched with Thai herbal extracts.

Moist healing environment: retains moisture to support optimal wound healing.

Exudate management: allows effective drainage of wound fluid through the porous mesh, helping reduce infection risk.

Flexible fit: highly conformable, suitable for curved or difficult-to-dress areas.

Botanical synergy: Centella asiatica supports anti-inflammatory activity and tissue regeneration, while Aloe vera supports healing and hydration.

More Information

Dent Booster: Desensitizing Gel for Tooth Sensitivity

Asst. Prof. Dr.Khrongkhwan Akkarachaneeyakorn
Affiliated Organization: Kasetsart University

Key Highlights

- Amorphous Calcium Phosphate (ACP) nanoparticles deeply seal open dentinal tubules
- Rapid relief from tooth sensitivity from the very first use



Amorphous Calcium Phosphate nanoparticles



Seals dentinal tubules



Promotes enamel remineralization

Enamel remineralization: strengthens tooth structure and reduces the risk of erosion and decay.

Proven and recognized: developed from peer-reviewed dental biomaterial research and recognized with international innovation awards for safety and efficacy.

Convenient home care: applied directly to sensitive areas without irritating oral tissues or affecting daily eating habits, ideal as preventive care alongside professional dental treatment.

More Information

Artificial Tears 0.28% Sodium Hyaluronate: VISMAX

Dr. Sujintana Tantaterdtham, M.D.
Affiliated Organization: **Ultra Medica Co., Ltd.**

Key Highlights

- 0.28% sodium hyaluronate eye drops for dry eye, with long-lasting corneal adhesion
- Clinically more effective than 0.18% sodium hyaluronate in corneal healing, visual acuity, and dry-eye relief
- Preservative-free and Thai FDA-approved



Long-lasting corneal adhesion



Preservative-free formula



Sterile single-dose vials

Ocular surface care: provides anti-inflammatory effects and protects corneal epithelial cells; its hypo-osmolar formula promotes natural corneal wound healing and restores tear film balance, with minimal irritation or tissue damage.

Approved uses: dry eyes, recovery after cataract and LASIK surgery, adjunct therapy for glaucoma, and relief of eye strain from dust, heat, air conditioning, prolonged screen use, or contact lenses.

More Information

Hygienic packaging: sterile blow-fill-seal (BFS) single-dose vials (0.5 mL), in boxes of 20 or 30.

Recognition: listed in the Thailand Innovation List, with SME-GP and Made in Thailand (MiT) certification.

Digital support: includes an online dry-eye self-assessment app, offering a comprehensive and affordable eye care solution.

BluRibbon & BluGel: Wound Dressing Patch and Medical Gel

Pharmacist Adisorn Apasuthirat
Affiliated Organization: **Novatech Healthcare Co., Ltd.**

Key Highlights

- Wound dressing sheet (BluRibbon) and gel (BluGel) using Nanobiocellulose and Blue Nano Silver
- For chronic, hard-to-heal wounds such as diabetic ulcers, pressure sores, and burns
- The only dressing that changes color when its silver is fully released – a visual cue to replace it



Nanobiocellulose and Blue Nano Silver



For chronic and hard-to-heal wounds



Color-change replacement indicator

Flexible formats:
the dressing sheet and conformable gel suit different wound shapes, helping accelerate healing and fight infection.

Awards:
National Innovation Award (Social Category, 2012) and the Gold Medal - The Grand Prix at the 40th International Exhibition of Inventions of Geneva (2012).

More Information

Cassava-based Medical Nutrition: Complete Nutrition Formula from Thai Cassava Starch

Asst. Prof. Dr.Supat Chaiyakul
Affiliated Organization: Mahidol University





Key Highlights

- Complete medical nutrition formula made from modified Thai cassava starch
- Low glycemic index: in-vitro GI values of 32.4 and 36.9
- 97.68 kcal per 100 mL, 250 mL per serving



Modified cassava starch



Complete nutrition formula



GI values of 32.4-36.9

Who it is for: individuals without diabetes, patients in recovery, people with malnutrition, and older adults with increased energy needs.

Facebook: Banpong Tapioca - Texture

Starch modification and formulation: both the formula and the production process were developed, using modified cassava starch as the carbohydrate source to replace imported starches, with optimized texture and digestibility.

More Information



Wellness & Longevity

Young Rice Jelly

Asst. Prof. Dr.Wasana Phanurak
Affiliated Organization: **Nakhon Ratchasima Rajabhat University**







Key Highlights

- Naturally sweet with no added sugar
- Boosts energy and helps address loss of appetite



Natural sweetness



No added sugar



Chewy and easy to eat

A healthy snack designed for picky-eating children and adults, with a soft, chewy texture that makes it easy and enjoyable to eat.

More Information

Effervescent Turmeric Extract Powder

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: **Mae Fah Luang University**





Key Highlights

- β-Cyclodextrin technology significantly improves curcumin solubility and absorption
- Laboratory studies show selective inhibition of colon cancer cells without toxicity to healthy cells



Enhanced curcumin solubility



Selective inhibition of colon cancer cells



Safety confirmed

Safety validated: acute toxicity testing under OECD Guideline 420 confirms its safety for consumption.

High-value health innovation: strong potential for commercialization as a premium herbal nutraceutical.

More Information

Red Herbal Pain Relief Oil

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: **Mae Fah Luang University**



Key Highlights

- Blend of methyl salicylate, Phlai essential oil, cooling aromatic compounds, and coconut oil
- Fast-acting relief for muscle pain, inflammation, spasms, and stiffness



Relieves muscle pain



Phlai and coconut oils



Helps reduce inflammation and spasms

A distinctive traditional red formula with trilingual labeling that supports wellness tourism, drives export opportunities, and adds value to native Thai herbal ingredients.

More Information

Curcumin-Zinc Skincare for Dark Spots and Melasma

Assoc. Prof. Dr.Nisakorn Saewan
Affiliated Organization: **Mae Fah Luang University**



Key Highlights

- Curcumin-Zinc Complex boosts Thai turmeric's stability, skin delivery, and anti-melanin efficacy
- Clinically tested: visible reduction of melasma and dark spots within 4 weeks



Curcumin-Zinc Complex



Reduces melasma and dark spots



Clinical results within 4 weeks

High economic and industrial impact: lowers active ingredient costs and reduces reliance on imported cosmetic ingredients.

Value-added Thai herbal innovation: transforms Thai herbal resources into high-value, competitive commercial skincare products.

More Information

Ready-to-Drink Ginger Beverage Fortified with L-Theanine

Kulraphat Wachirasiri, Ph.D.

Affiliated Organization: **Thailand Institute of Scientific and Technological Research (TISTR)**



Key Highlights

- Functional drink combining natural ginger extract and L-theanine to promote relaxation and well-being
- Helps reduce stress and supports a calm yet alert, focused mind
- Mild, non-spicy taste, suitable for all genders and ages

Natural ginger with L-theanine



Promotes relaxation and stress relief



Mild, ready-to-drink beverage

Product details: 200 mL bottle, 60 THB, for daily drinking.

Benefits of ginger: rich in vitamins and minerals; eases indigestion and bloating, aids nutrient absorption, and its 50+ antioxidants help protect brain cells.

Mood and brain support: gingerols and shogaols help ease stress, anxiety, and headaches, and are linked to better sleep and brain protection.

How L-theanine works: boosts calming alpha brain waves and lowers tension-linked beta waves; about 200–250 mg/day is considered safe.

More Information

Chiwadi Organic Coconut Blossom Seasoning Sauce

Saraphi Yuadyong
Affiliated Organization: **Chiwadi Products Co., Ltd**



Key Highlights

- Plant-based seasoning sauce from coconut blossom nectar, free from enzymatic or acid protein hydrolysis
- Only 190 mg sodium per serving, versus 400-600 mg in typical seasoning sauces
- Natural umami taste with no animal-derived ingredients or synthetic flavor enhancers

Plant-based umami flavor

Free from animal-derived ingredients

190 mg sodium per serving

Safety and purity: avoids allergenic peptides and contaminants such as 3-MCPD found in conventional fish sauces.

Designed amino acid profile: enhances coconut nectar's natural umami, true to Thai cuisine.

Natural mineral blend: mild saltiness from potassium, phosphorus, magnesium, manganese, and calcium.

Chiwadi Organic Coconut Blossom Seasoning Sauce blends traditional Thai taste with modern nutrition science.

More Information

Super Concentrate Gen. II: Multi-Dimensional Anti-Aging Peptide Serum

Assoc. Prof. Dr.Rawiwan Charoensup

Affiliated Organization: **Mahidol University and Siam Bioscience Co., Ltd.**





Key Highlights

- Thai skin-longevity serum powered by bio-signal peptides and multi-growth factor technology (EGF)
- Visible improvement in fine lines within 7 days



Bioactive peptides



Multi-Growth Factor Technology



Testing results within 7 days

Biotechnology-based formula: developed through peptide-based skin science to support skin renewal, improve the appearance of aging skin, and enhance skin vitality at the cellular level.

Bio-signal peptides: peptide-based signaling molecules support the skin's natural repair and regeneration processes.

Multi-growth factors: EGF (Epidermal Growth Factor) combined with multiple growth-factor components supports skin recovery, renewal, and elasticity.

Deep cellular support: high-potency bioactive ingredients help reinforce the skin barrier for healthier, more resilient-looking skin.

More Information

Wang Som Sa Model: From Thai Biodiversity to a Zero-Waste Wellness Ecosystem

Asst. Prof. Dr.Chanai Noisaeng
Affiliated Organization: Rajamangala University of Technology Thanyaburi



Key Highlights

- Transforms Som Sa (bitter orange) into high-value wellness, beauty, spa, and functional food products
- 100% zero-waste model: every part of the fruit is upcycled



Advanced bio-extraction innovation



Uses every part of Som Sa



Zero-waste model

Zero-waste in practice: outer peel (essential oils and cosmetic ingredients), inner peel (pectin-based prebiotic ingredients), seeds (cold-pressed skincare oil), leaves (aromatic herbal tea), and juice (premium functional beverages).

Advanced bio-extraction: hydrodistillation, ultrasound-assisted, solvent, microwave-assisted, and supercritical fluid extraction optimize bioactive yield and stability.

More Information

Research-backed extracts: high-value phytochemical extracts with antioxidant, antimicrobial, metabolic, and wellness-related bioactivities.

Community enterprise: developed in collaboration with local community enterprises.

Ready-to-Use *Bacopa monnieri* (Brahmi) Extract in Powder Form

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: **Naresuan University**



Key Highlights

- Ready-to-use, standardized Bacopa monnieri (Brahmi) extract powder
- Produced in a GMP-certified pilot plant with herbal raw material quality certification from the Thai FDA

Powdered Bacopa monnieri extract

GMP standard

Thai FDA-certified raw-material quality

Production site: the Herbal Extract and Product Pilot Plant, Faculty of Pharmacy, Naresuan University.

Target industries: health, dietary supplement, and elderly products, both domestically and internationally, with applications in cognitive support, memory enhancement, and wellness formulations.

More Information

Export ready: bridges academic research with commercial scalability, offering a safe, reliable, compliant herbal raw material for local and international markets.

Acanthus MicroCare Hair Tonic: Thai Herbal Microemulsion Innovation for Hair & Scalp Wellness

Asst. Prof. Dr.Suradwadee Thungmungmee
Affiliated Organization: Rajamangala University of Technology Thanyaburi

Key Highlights

- Research-based hair tonic powered by Acanthus, a Thai medicinal plant, to support healthier hair roots and help reduce and prevent hair fall
- Microemulsion technology with nanoscale droplets for better delivery of herbal actives to the scalp
- Lab-proven: Acanthus leaf extract (12.5 µg/mL) boosted hair-root cell growth about 1.22 times vs. control, with no cytotoxicity



Acanthus ebracteatus extract



Nano-scale microemulsion



Low-energy production process

Stable formulation: the microemulsion system maintains high formulation stability while enhancing the effectiveness of active compounds associated with hair-root growth.

Sustainable production: uses a low-energy production process, with packaging designed to improve stability and support reuse.

More Information

Contact: suradwadee_t@rmutt.ac.th

Plant-Based Avocado Salad Dressing Enriched with Macadamia Milk

Asst. Prof. Dr. Janjira Phakawan
Affiliated Organization: **Rajamangala University of Technology Thanyaburi**



Key Highlights

- Plant-based dressing made with Hass avocado and macadamia milk – creamy and smooth, with no eggs, dairy, or added starch
- Rich in healthy fats: unsaturated fatty acids make up 88.91% of total fatty acids
- Natural Omega-3, 6, 7, and 9



Plant-based, dairy-free, and egg-free



Creamy texture



Industrial-scale production

Plant-Based Emulsion System: combines hydrocolloids and plant-based emulsifiers in optimized proportions to create a highly stable texture that resists oil-water separation.

Balanced lipid profile: contains both MUFA and PUFA, derived from avocado, macadamia, and plant oils.

Inclusive choice: a nutrient-dense option for vegans, lactose-intolerant individuals, egg-allergic consumers, and health-conscious diets.

Added value: creates value for Thai agricultural ingredients and provides a platform for new healthy food products for domestic and international markets.

More Information

Soy Milk Powder Blended with Vietnamese Coriander Extract and Mulberry Leaf (MUVITA)

Prof. Jintanaporn Watthanathorn
Affiliated Organization: Khon Kaen University





Key Highlights

- Soy milk powder blended with Vietnamese coriander extract and mulberry leaf, in a convenient daily sachet
- One sachet a day for 30 days reduces stress, anxiety, and depression and improves sleep quality
- Enhances working memory and cognitive speed



Vietnamese coriander and mulberry leaf



30-day consumption data



Supports sleep and memory

Supports bone health by balancing bone metabolism – increasing bone formation while reducing bone resorption.

More Information

PineCoBeads: Biodegradable Beads from Pineapple Rhizome Starch

Assoc. Prof. Dr.Taweechai Amornsakchai
Affiliated Organization: **Mahidol University**

Key Highlights

- 100% biodegradable beads made from upcycled pineapple rhizome starch
- A sustainable, microplastic-free alternative to plastic beads
- Smooth, spherical particles for gentle exfoliation



Upcycled pineapple stem starch



100% biodegradable



Gentle exfoliation

Circular beauty: turns an underutilized agricultural by-product into a functional cosmetic ingredient, helping reduce waste, create added value, and support more sustainable beauty formulations.

Formulation-ready: water-resistant with good stability in water-based formulations, suitable for rinse-off skincare and cleansing products.

More Information

BioBlend+ Calcium: Fish-Derived Calcium Innovation for Bone Longevity

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: **Mae Fah Luang University**



Key Highlights

- Calcium supplement made from MCHC (Microcrystalline Hydroxyapatite) from Maejo Siam hybrid catfish bone
- Ca ratio ≈ 1.67, closely matching human bone
- Evaluated in human clinical research, including studies in menopausal women

Fish-derived MCHC calcium

Calcium ratio similar to human bone

Naturally occurring calcium and phosphorus

Bone-like mineral matrix: provides naturally occurring calcium and phosphorus in a hydroxyapatite matrix, designed to support bone strength and musculoskeletal health with good absorption.

Added value: transforms fish-bone resources into a higher-value nutritional ingredient through bio-innovation.

More Information

Sustainable Highland Tea Innovation

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: **Department of Agriculture**



Key Highlights

- Transforms Northern Thai tea into higher-value wellness products through smart farming, advanced processing, and market development
- Five market-ready teas: matcha, natural Thai tea, plum-flavored black tea, vanilla GABA tea, and Thai genmaicha
- Empowers 120 farmers across 6 community enterprises



120 participating farmers



Five value-added tea products



Tea tourism and training

Whole value chain: strengthens every step from farm management and quality improvement to processing, product development, and market access.

Advanced processing: research-backed processing optimizes active phytonutrients and GABA content, targeting the premium wellness market; selected products show significant increases in selling value compared with their previous forms.

Beyond tea products: tea tourism, tea brewing experiences, value-added processing, and Tea Guide training create new income opportunities for local communities.

More Information

Jasmine Rice Young Leaf Tea

Asst. Prof. Dr.Supat Chaiyakul
Affiliated Organization: **Mahidol University**





Key Highlights

- Transforms young leaves of Thai Hom Mali Rice 105 into high-value functional tea and food ingredients
- Creates new wellness products and additional income for farmers



Integrated cultivation and harvesting



Bioactive-compound analysis



High-quality standardized raw materials

Standardized cultivation system: an integrated cultivation, harvesting, and field management system with raw material quality control ensures consistent, high-quality raw materials.

Bioactive and aroma profiling: key bioactive compounds, including natural antioxidants and the signature aromatic compounds of Hom Mali rice, are analyzed.

Commercial-ready ingredient: a standardized botanical raw material suitable for further development into wellness, beverage, and functional food products.

More Information

Rice Panicle Bioactive Innovation: From Thai Rice to Advanced Skin Wellness

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: **Mae Fah Luang University**





Key Highlights

- Transforms underutilized Thai rice panicles into high-value bioactive ingredients for cosmeceutical and wellness use
- Antioxidant activity plus inhibition of tyrosinase, elastase, and collagenase
- Naturally contains Gamma Oryzanol and Ferulic Acid



Rice panicle extract



Gamma Oryzanol and Ferulic Acid



Reported non-irritating to skin

Anti-aging action: inhibiting these enzymes helps prevent skin hyperpigmentation and collagen breakdown.

Line: @chailaioffice

Stability and safety: a specialized extraction process yields an extract with good physical and chemical stability, reported to be non-irritating to the skin, supporting its use in advanced skincare formulations.

More Information

Arabica Coffee with Yamabushitake (Lion's Mane) Extract

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: Rajamangala University of Technology Lanna



Key Highlights

- Functional coffee for cognitive wellness
- Locally grown Arabica coffee from Nan Province enriched with Lion's Mane mushroom (Hericium erinaceus) extract



Nan-grown Arabica coffee



Lion's Mane mushroom extract



Supports cognitive wellness

Lion's Mane is a functional mushroom widely recognized in traditional wellness practices and increasingly studied for its potential role in supporting focus, cognitive function, and the nervous system. The blend keeps the rich, familiar flavor and aroma of Thai Arabica coffee in an easy-to-consume daily format, designed to support cognitive wellness and everyday vitality.

More Information

Thai Cacao Nibs: Naturally Functional Cacao

Asst. Prof. Dr.Chinanart Wittayapraphakorn
Affiliated Organization: **Rajamangala University of Technology Lanna**





Key Highlights

- Premium roasted cacao nibs from Thai-grown cacao
- Minimally processed to retain naturally occurring bioactive compounds
- Rich in antioxidants and plant bioactives associated with cardiovascular and overall wellness

Roasted cocoa nibs

Naturally occurring antioxidants

Developed with local producers

Improved production and processing: developed with local producers, using optimized fermentation and roasting that protect nutritional quality while enhancing the authentic flavor of Thai cocoa.

Natural functional compounds: contains polyphenols and flavonoids associated with healthy blood flow and overall vitality.

Versatile clean-label use: a nutrient-dense superfood for direct snacking, functional food formulations, and specialty confectionery.

More Information

SimpleSense : High-Protein, Low-GI Ready-to-Eat Puree for Special Care Groups

Asst. Prof. Dr.Phimnipa Hiransorn
Affiliated Organization: Khon Kaen University







Key Highlights

- Ready-to-eat high-protein puree for patients, the elderly, and people with swallowing difficulties (dysphagia)
- Safe-swallow texture meeting IDDSI Level 4, clinically tested for swallowing safety
- High protein, low GI, low sodium and phosphorus, dairy-free



High-protein ready-to-eat meal



IDDSI Level 4 texture



Dairy-free

Developed by experts: food technologists and rehabilitation medicine specialists from Khon Kaen University.

Complete nutrition: balanced, high energy and nutrient density even in small portions.

Appetite appeal: refreshing flavors scored significantly higher in palatability than reference foods in sensory evaluation.

Convenient storage: 12-month shelf life at room temperature, no refrigeration required.

More Information

PROTINOS – High-Protein Noodle Innovation for Stronger Muscles

Dr.Peerapong Ngamnikom and Dr.Siriluck Surin
Affiliated Organization: Rajamangala University of Technology Thanyaburi



Key Highlights

- High-protein dried noodles: 20-30 g protein per serving (equal to 5-8 egg whites)
- 4,000-10,000 mg of BCAAs to support muscle growth and prevent muscle loss
- Made from egg white and soy protein



High-protein noodles



20-30 g protein per serving



4,000-10,000 mg BCAAs

Developed from scientific research, PROTINOS uses enzyme technology to create a smooth, chewy texture similar to traditional noodles. It offers a convenient, protein-rich meal option with variety for active people who want to build and maintain muscle, support recovery, and stay strong and healthy.

More Information

ALTERMED: Biotica Postbiotics Relief Mask

Assoc. Prof. Dr.Rawiwan Charoensup
Affiliated Organization: **Mae Fah Luang University**

Key Highlights

- Facial mask powered by postbiotic technology
- Calms sensitive skin and visibly reduces signs of irritation
- Strengthens the skin's natural protective barrier



Postbiotic facial mask



Soothes visible irritation



Supports the skin barrier

Skin recovery: supports skin microbiome balance and everyday skin recovery.

Barrier protection: reinforces the skin barrier against environmental stressors and moisture loss.

Preventive and restorative approach: combines postbiotic science with barrier-focused skincare for daily skin wellness.

Sheet mask format: a skin-conforming sheet designed for absorption, hydration, and gentle use on delicate skin.

More Information

Functional Aromatic Blends for Relaxation, Refreshment, and Rest

Dr. Orapreeya Boonrat
Affiliated Organization: **Prince of Songkla University**

Key Highlights

- Essential oil roll-ons for daily wellness, from daytime refreshment to evening relaxation and sleep
- Created with Functional Fragrance Design and Synergistic Blending
- Pilot human testing with EEG brain-wave monitoring



Two wellness blends



Molecularly informed fragrance design



Balanced, distinctive scent profile

Formulation science: each blend considers chemical composition, molecular volatility, and each ingredient's role across Top, Heart, and Base notes, creating balanced, smooth scent transitions with a distinctive identity tailored to a specific wellness purpose.

Minty Reset (Refresh • Reset • Recharge): for daytime freshness and focus; peppermint, rosemary, lemon, lavender, and vetiver in a cool herbal-citrus profile.

Sleep Relax Essential Oil Roller: for bedtime relaxation; bergamot, vetiver, sandalwood, ylang-ylang, jasmine, and Atlas cedar.

Mind Relax (Breathe • Unwind • Rest): for evening unwinding; lavender, frankincense, bergamot, and vetiver in a soft floral, resinous, citrus, and woody profile.

EEG results: reported data show changes in relative brain-wave patterns across different frequency bands during essential oil exposure.

More Information

SORA Nano Bubbles: Premium Micro/Nano Bubble Skincare

Assoc. Prof. Dr.Sorapong Pavasupree

Affiliated Organization: **Rajamangala University of Technology Thanyaburi**



Key Highlights

- Micro/nano bubble technology: ultra-fine oxygen bubbles smaller than 1 micron in water
- Deeper, longer-lasting hydration in a lightweight, water-based texture
- 100% Clean Beauty: no added color, fragrance, preservatives, or alcohol



Micro/nano oxygen bubbles



Deep, long-lasting hydration



100% Clean Beauty formula

Validated technology: nano bubble size and dispersion verified with NanoSight NS300, with a reported zeta potential of -17.05 mV.

Gentle formula: designed to be suitable even for sensitive skin.

Commercial reach: a premium skincare platform with commercial applications in Thailand and Japan.

More Information

KORA Phyto-Element: Thai Elemental Wellness Powered by Advanced Phytosome Technology

Assoc. Prof. Dr.Rawiwan Charoensup

Affiliated Organization: **Rajamangala University of Technology Thanyaburi**



Key Highlights

- Personalized supplements based on the Thai Traditional Medicine Four-Element concept: Earth • Water • Wind • Fire
- Phytosome Technology improves absorption and bioavailability of plant bioactives



Four-Element personalized supplements



Phytosome-enhanced absorption



Thai wisdom meets modern science

Fire Element: skin nourishment and internal refreshment, with selected Thai botanicals traditionally linked to internal balance and skin wellness.

Wind Element: supports relaxation, comfort, and overall daily well-being.

Water Element: daily health protection and vitality, with pineapple, orange, and tomato.

Earth Element: holistic nourishment and daily strength, with peanut, cabbage, and mangosteen.

Science meets tradition: combines Thai traditional wisdom with pharmaceutical formulation science and molecular-level research.

More Information